Comparing Markov Chain Monte Carlo Samplers for Bayesian Linear Regression: Convergence, Efficiency and Predictive Calibration in Cloud CPU Forecasting

Elad Dagmi & Shaked Mizrahi

Advanced Methods in Machine Learning

August 2026

Institution: [to be completed]

Course: Advanced Methods in Machine Learning

Lecturer: Dr. Boaz Tamir

Student IDs: [to be completed]

Source code and data instructions: https://github.com/eladagmi24/mcmc-sampling-project

Abstract

Markov chain Monte Carlo (MCMC) makes Bayesian inference possible when the posterior has no closed form, but its reliability rests on diagnostics that are easy to misread. This study compares five samplers, implemented from scratch, on Bayesian linear regression for forecasting virtual-machine CPU utilisation 5 minutes ahead, using the Bitbrains GWA-T-12 datacenter traces [1]. The samplers are random-walk, adaptive and Fisher-preconditioned Metropolis, Gibbs sampling and Hamiltonian Monte Carlo. Observations are split chronologically, with an embargo against feature windows crossing boundaries and scaling estimated on training data alone. Convergence is assessed with rank-normalised and folded $\hat{R}$ and with bulk and tail effective sample size (ESS), computed jointly across 4 dispersed chains over all 12 model parameters [2]. Gibbs and Hamiltonian Monte Carlo converge; preconditioned Metropolis converges only with a longer run. Plain and adaptive Metropolis do not, reaching $\hat{R}$ 4.53 and 3.48, with posterior means wrong by 1.36 and 1.44 against a deterministic quadrature reference. The cause is posterior anisotropy, condition number 314. Preconditioning repairs the geometry: at 40,000 draws it converges at bulk ESS 2117, against 4.2 for plain Metropolis. The converged samplers reproduce the quadrature reference to within 0.0010. Test residuals have an excess kurtosis of 111, so a Gaussian likelihood yields posterior predictive intervals that are far too wide, nominal 50% intervals covering 94.7%. A Student-t likelihood, with degrees of freedom selected on validation data, reduces this to 74.1% and cuts the median absolute error from 0.210 to 0.072 CPU percentage points, though coverage remains well above nominal.

Table of Contents

1. Introduction

Bayesian inference requires the posterior distribution of the model parameters given the data. For most models the normalising constant cannot be computed, so the posterior is explored by simulation. MCMC constructs a Markov chain whose stationary distribution is the target posterior [3, 4], which turns inference into a simulation problem and, with it, into a question of whether the simulation has run long enough.

The application is forecasting the CPU utilisation of virtual machines in a production datacenter [1]. A point forecast is of limited operational value; capacity planning and overload detection need a calibrated statement of uncertainty, which is what a posterior predictive distribution provides.

This study addresses three research questions:

- RQ1. Which samplers actually converge on this posterior, and what property of the posterior explains the differences between them?
- RQ2. Among the samplers that converge, which is most efficient per unit of computation, measured by effective sample size per second and, for gradient methods, per gradient evaluation?
- RQ3. Are the resulting posterior predictive intervals calibrated on held-out data, and if not, what is responsible?

Five samplers target the same Gaussian posterior: random-walk Metropolis-Hastings, adaptive Metropolis [5], Metropolis preconditioned by the observed Fisher information, Gibbs sampling [6] and Hamiltonian Monte Carlo [7]. A sixth sampler, a Gibbs sampler for Student-t regression, targets a different model and is therefore discussed separately, in Section 6, rather than being compared with the five on convergence or efficiency.

2. Theoretical Background

2.1 Model and Priors

The regression model for the standardised target is

$$y \mid X, \beta, \sigma^2 \sim N(X\beta, \sigma^2 I\_n)$$

The two priors are specified independently of one another:

$$\beta \sim N(0, \tau^2 I\_p), \quad \tau^2 = 10$$

$$1/\sigma^2 \sim Gamma(a0 = 2, rate = b0 = 1), \ equivalently\ \ \sigma^2 \sim Inverse\text{-}Gamma(a0 = 2, scale = b0 = 1)$$

The precision $1/\sigma^2$ follows a Gamma in shape-rate form; inverting gives σ^2 an Inverse-Gamma in shape-scale form, with density proportional to (σ^2)^(-a0-1) exp(-b0 / σ^2). The Gamma distributions appearing in the Student-t sampler of Section 2.6 are likewise in shape-rate form.

This matters in implementation because NumPy draws Gamma variates with a scale argument, so every rate is passed as its reciprocal.

Note that the prior on β does not scale with σ^2. The joint prior is therefore not the conjugate Normal-Inverse-Gamma, and the joint posterior has no standard closed form, although both full conditionals remain standard. This is what makes Gibbs sampling available while still leaving something for the other samplers to do.

2.2 Metropolis-Hastings

A proposal θ' is accepted with probability

$$\alpha = min(1, [\pi(\theta')\, q(\theta \,/\, \theta')] \,/\, [\pi(\theta)\, q(\theta' \,/\, \theta)])$$

For a symmetric random-walk proposal the q terms cancel. The method needs only pointwise evaluation of an unnormalised density, which makes it general, but an uninformed random walk explores slowly. For a d-dimensional Gaussian target the asymptotically optimal acceptance rate is about 23.4 percent [8].

2.3 Adaptive and Preconditioned Metropolis

A single scalar step size implicitly assumes a spherical posterior. When the posterior is anisotropic the step must fit the narrowest direction while the chain has to traverse the widest, and the number of steps required grows with the condition number of the posterior covariance. Two remedies are examined.

Adaptive Metropolis [5] estimates the proposal covariance from the chain history and proposes from $N(\theta, (2.38^2 / d)\, C)$. Preconditioned Metropolis instead takes the metric from the observed Fisher information, evaluated at the least-squares residual variance $\hat{\sigma}^2$:

$$(X'X \,/\, \hat{\sigma}^2 + I \,/\, \tau^2)^{-1} = \hat{\sigma}^2 (X'X + \hat{\sigma}^2 I \,/\, \tau^2)^{-1}$$

The right-hand form is the one implemented. It requires a single least-squares fit and no conjugacy. A Robbins-Monro rule additionally tunes a global scale toward the optimal acceptance rate, and the empirical covariance of the second half of burn-in refines the metric. Both adaptations stop before the retained draws begin, so the retained chain is a homogeneous Markov chain and no diminishing-adaptation argument is needed.

2.4 Gibbs Sampling

Both full conditionals are standard:

$$\beta \,/\, \sigma^2, y \sim N(\Sigma X'y \,/\, \sigma^2, \Sigma),\ \Sigma = (X'X \,/\, \sigma^2 + I \,/\, \tau^2)^{-1}$$

$$\sigma^2 \,/\, \beta, y \sim Inverse\text{-}Gamma(a0 + n/2,\ b0 + ||y - X\beta||^2 \,/\, 2)$$

Every draw comes from an exact conditional, so nothing is ever rejected. There is no accept-reject step to measure, so no acceptance rate is defined for either Gibbs sampler and the tables record it as not applicable rather than as one. Quoting one would invite comparison with the tuned rates of the Metropolis samplers, where the number reflects how well the proposal is matched to the target; here it would reflect only the structure of the algorithm. The comparison in Section 5 therefore rests on $\hat{R}$, ESS and runtime.

2.5 Hamiltonian Monte Carlo

HMC augments the parameter space with a momentum variable and simulates Hamiltonian dynamics with the leapfrog integrator, which is symplectic and reversible, so detailed balance holds once the accept-reject step corrects the integrator error [7].

$$H(q, p) = -\log \pi(q) + p'p / 2$$

Because σ^2 must stay positive, HMC samples the unconstrained parameter $\psi = \log(\sigma^2)$. The change of variables contributes a Jacobian $|d\sigma^2 / d\psi| = \exp(\psi) = \sigma^2$, so the term $\log(\sigma^2)$ is added to the log posterior. The same transformation and the same Jacobian term are used by every sampler that works on the unconstrained scale, namely both Metropolis variants, HMC and the external emcee reference; Gibbs works directly on σ^2 and needs no Jacobian. The gradient used by the leapfrog integrator is the analytic gradient of this transformed log posterior.

The No-U-Turn Sampler, which tunes the trajectory length automatically, was not implemented or run in this study. Where trajectory length matters it is varied explicitly, in Section 5.6.

2.6 Student-t Regression as a Normal Scale Mixture

A Gaussian likelihood treats large residuals as essentially impossible, so when the data contain rare extreme observations the fitted noise variance inflates to accommodate them and every predictive interval widens. A Student-t likelihood tolerates them. Writing it as a scale mixture keeps every full conditional standard [9, 10]:

$$y_i \mid \beta, \sigma^2, w_i \sim N(x_i'\beta, \sigma^2 / w_i), \quad w_i \sim \mathrm{Gamma}(\nu/2, \nu/2)$$

Marginalising the weights w_i recovers a Student-t with ν degrees of freedom and scale σ. Conditional on the weights the model is a weighted Gaussian regression, so the sampler remains a tuning-free Gibbs sampler in which an observation with a large standardised residual receives a small weight and is automatically down-weighted.

2.7 Convergence Diagnostics

Notation. Each of the C sampled chains is split in half, giving $M = 2C$ split chains of length N, so the total number of draws is $S = M N$. Write W for the mean within-split-chain variance, B for the between-split-chain variance, ρ_t for the autocorrelation at lag t and τ for the integrated autocorrelation time.

Rank-normalised and folded $\hat{R}$.

The classical statistic [11] compares between-chain and within-chain variance, $\hat{R} = \sqrt{V / W}$ with $V = ((N-1)/N) W + B/N$. It assumes finite variance and is not invariant to monotone transformations. We therefore use the rank-normalised version [2]: draws are pooled, replaced by their ranks with ties averaged, and mapped through the inverse normal cdf with the Blom transform $z = \Phi^{-1}((r - 3/8)/(S - 1/4))$ before the classical formula is applied. The folded variant applies the same procedure to $|\theta - \text{median}(\theta)|$, which makes it sensitive to disagreement in scale rather than location. The reported value is the larger of the two. The threshold used throughout is 1.01.

Bulk and tail effective sample size.

ESS = S / τ, where $\tau = 1 + 2 \sum_{t\geq1} \rho_t$. The autocorrelations are estimated jointly across chains through $\rho_t = 1 - (W - \text{mean}_m[s_m^2 \rho_{t,m}]) / V$, so that between-chain variation enters the estimate, and the sum is truncated by Geyer's initial monotone positive sequence rule [12]. Bulk-ESS applies this to the rank-normalised draws and describes the centre of the distribution; tail-ESS is the smaller of the ESS values for the 5 percent and 95 percent quantile indicator series and describes the tails. Both are single numbers computed from all S draws jointly, never per-chain values that are then averaged.

Aggregation across parameters.

Each diagnostic is computed for one parameter at a time, for all 11 regression coefficients and the noise variance σ^2, 12 parameters in total. The single number reported for a sampler is the worst case: the maximum $\hat{R}$ and the minimum bulk and tail ESS over those parameters. Taking the worst rather than an average prevents one badly behaved coordinate from being hidden by well behaved ones. Efficiency per second uses the same minimum bulk ESS and the same total runtime, so the ratio is internally consistent.

Monte Carlo standard error.

MCSE is the posterior standard deviation divided by the square root of the joint bulk ESS, and expresses how precisely a posterior mean has been located by a finite run. It is reported only for chains that converged, because for a chain that has not converged the estimate is biased and its standard error understates the true error.

Implementation check.

All diagnostics are implemented from scratch. To confirm the implementation rather than assert it, the estimators were compared with ArviZ [13], the reference implementation of the same paper. For the Gibbs intercept our $\hat{R}$ is 0.99997 against ArviZ 0.99997, and our bulk ESS is 40111.4 against 40111.8. Across a set of synthetic stress cases including independent draws, a strongly autocorrelated series, offset chains and heavy-tailed draws, the two agreed to within 0.4 percent on ESS.

2.8 Computational Complexity

With n observations, p parameters, L leapfrog steps and T retained iterations:

Table 1. Per-iteration time and total space complexity of each sampler.

Sampler	Time per iteration	Space	Dominant cost
Metropolis-Hastings	$O(np)$	$O(np + Tp)$	one likelihood evaluation
Adaptive Metropolis	$O(np + p^2)$	$O(np + p^2 + Tp)$	likelihood and proposal

Sampler	Time per iteration	Space	Dominant cost
Preconditioned MH	$O(np + p^2)$	$O(np + p^2 + Tp)$	likelihood and proposal
Gibbs	$O(np + p^3)$	$O(np + p^2 + Tp)$	p by p solve for β
HMC	$O(L\,np)$	$O(np + Tp)$	L gradient evaluations
Student-t Gibbs	$O(np^2 + p^3)$	$O(np + p^2 + Tp)$	reweighted X'WX each sweep

Preconditioning adds only an $O(p^2)$ multiplication by a Cholesky factor, negligible beside the $O(np)$ likelihood, so the efficiency it buys is close to free. The Student-t sampler is the most expensive per sweep because the weights change every iteration, so X'WX cannot be cached. With $n = 3{,}000$ and $p = 11$, $np = 33{,}000$ dominates $p^3 = 1{,}331$.

3. Data and Experimental Design

3.1 Dataset

The Bitbrains GWA-T-12 traces [1] were collected from a managed hosting datacenter operated by Bitbrains IT Services in the Netherlands, and are distributed through the Grid Workloads Archive. The fastStorage trace records eleven telemetry channels for 1,250 virtual machines at five-minute resolution over roughly thirty days in August and September 2013. We use 48 machines.

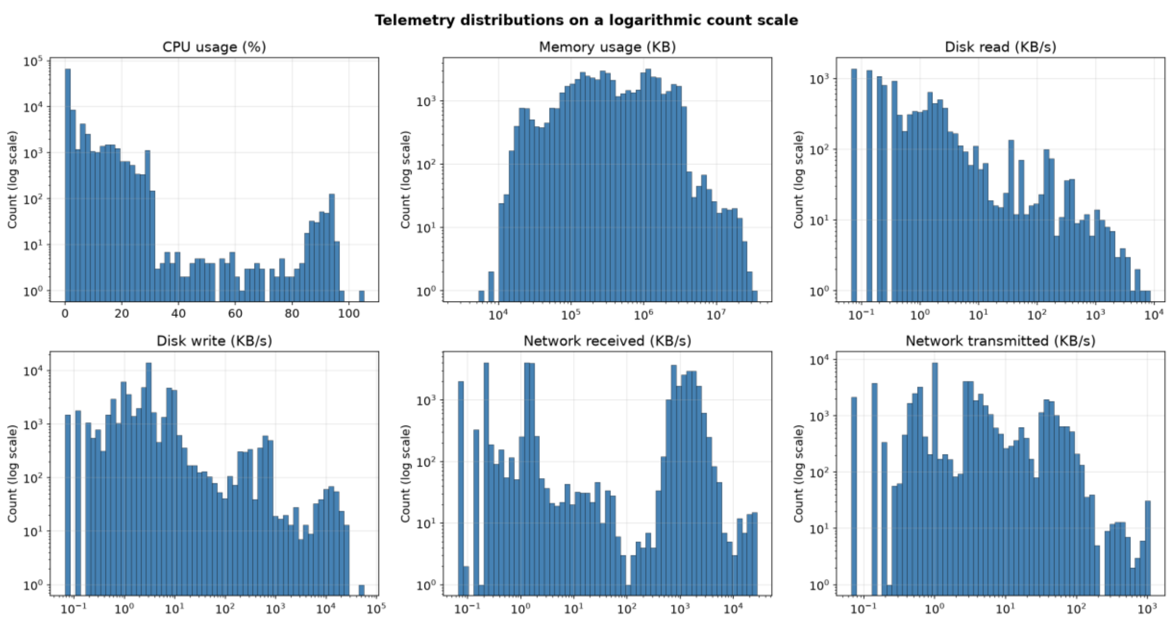

Figure 1. Distributions of the six telemetry channels used as predictors. Counts are on a logarithmic scale, and the five strictly positive channels also use a logarithmic horizontal axis, because on linear axes every distribution collapses into a single bar at the left edge.

3.2 Forecasting Task and Feature Construction

The task is one-step-ahead forecasting at a horizon of 5 minutes: the target is CPU utilisation at time t, and every predictor is a function of information available strictly before t. This is stated explicitly because it constrains the design. The contemporaneous memory, disk and network channels are not observable when the forecast is issued, so each is lagged by one interval. Had they been used at time t the exercise would have been nowcasting rather than forecasting.

The 10 predictors, plus an intercept, are:

- Exogenous load at t-1: memory usage, disk read and write throughput, network received and transmitted throughput.
- Autoregressive terms: CPU utilisation at t-1, t-2 and t-3.
- Rolling statistics of CPU utilisation over the six intervals ending at t-1: mean and standard deviation.

3.3 Chronological Splitting and Leakage Control

Observations from all machines are ordered by timestamp and divided chronologically into training, validation and test periods in proportions 60, 20 and 20 percent. Three separate measures guard against leakage.

- Feature windows may not straddle a boundary. Each row depends on the six intervals preceding it, so any row whose earliest required observation falls in an earlier split is discarded. This embargo removed 632 rows.
- All scaling constants, the predictor means and standard deviations and those of the target, are estimated on the training rows only and then applied unchanged to validation and test.
- The validation split is used to choose the likelihood and its degrees of freedom. The test split is used once, for the final numbers in Section 6, and for nothing else.

Table 2. Split sizes after the embargo.

Split	Rows	Role
Training	3,000	Posterior sampling
Validation	688	Likelihood and degrees-of-freedom selection
Test	680	Final reported performance only
Embargoed	632	Discarded to prevent window crossing

One qualification should be stated plainly. The same machine appears in more than one split, at different times. For time-series forecasting this is the intended design, since the point is to predict a machine's future from its past, and the embargo ensures no individual prediction uses information from across a boundary. It does mean the splits are not independent in the sense of disjoint machines, so the results describe forecasting for machines already observed, not generalisation to unseen machines.

3.4 Sample Size

Of roughly 468,341 raw records available across the selected machines, the study uses the earliest 5,000 engineered observations. The restriction is computational rather than statistical. Each sampler is run with 4 chains, 10,000 draws per chain and 3 independent repeats, and every sampler carries a per-iteration cost that grows linearly in n, from $O(np)$ for Metropolis-Hastings through $O(np + p^3)$ for Gibbs and $O(Lnp)$ for HMC to $O(np^2 + p^3)$ for the Student-t sampler, as Table 1 sets out; the full experiment already requires several hundred thousand likelihood or gradient evaluations at $n = 3{,}000$. Using the whole trace would multiply that by roughly two orders of magnitude without changing what the experiment is designed to measure, which is the relative behaviour of the samplers on a fixed posterior. The consequence is that the test split contains only 680 rows, so the coverage estimates in Section 6 carry appreciable uncertainty, which is why they are reported with bootstrap intervals.

3.5 Pooling Across Machines

All machines share a single coefficient vector. This is a modelling assumption, and the residuals show it is only approximately right: across the 48 machines with at least ten training rows, the per-machine residual standard deviation ranges from 0.000 to 1.575 with a median of 0.049, against a pooled value of 0.349. A machine at the upper end therefore has noise several times larger than one at the lower end, and a single σ^2 splits the difference, over-stating uncertainty for quiet machines and under-stating it for volatile ones. A hierarchical model with per-machine coefficients and variances is the natural remedy and is left to further work; it would also make the comparison between Gibbs and HMC considerably more interesting, since conjugacy alone no longer suffices there.

4. Methodology

All samplers are implemented from scratch in NumPy and SciPy. No probabilistic programming library is used inside any implementation; emcee appears only as an external reference in Section 5.5, and ArviZ only to verify the diagnostic code.

Each sampler is run with 4 chains from overdispersed starting points, drawn as $\beta \sim N(0, 2^2)$ and $\log \sigma^2 \sim N(0, 1.5^2)$, which are far wider than the posterior. Overdispersed starts are what split $\hat{R}$ assumes: if every chain began near the mode, agreement between chains would demonstrate little. Each chain retains 10,000 draws after 2,000 burn-in iterations, and the whole procedure is repeated 3 times with different seeds so that run-to-run variability in runtime and ESS can be reported.

Table 3. Tuning parameters. Gibbs samplers have none.

Sampler	Tuning parameters	Setting
Metropolis-Hastings	proposal scales for β and $\log \sigma^2$	0.001, 0.05
Adaptive Metropolis	adaptation interval, scaling	200 iterations, $2.38^2/d$
Preconditioned MH	target acceptance, adaptation interval	0.234, 100

Sampler	Tuning parameters	Setting
Gibbs	none	not applicable
HMC	step size, leapfrog steps	0.002, 15
Student-t Gibbs	degrees of freedom ν	selected on validation

5. Results: Convergence and Efficiency

5.1 Posterior Geometry

Because the conditional posterior of β given σ^2 is Gaussian, its covariance is available in closed form. Evaluated at the fitted noise level it has condition number 313.8 and a largest absolute correlation between coefficients of 0.686, with standard deviations along the extreme directions of 0.0028 and 0.0499. The correlation is a direct consequence of the design: three consecutive CPU lags and a rolling mean of the same series measure nearly the same quantity. A single isotropic step size must be small enough to be accepted along the narrow direction while the chain has to travel roughly 18 times further to cross the wide one.

5.2 Convergence

Table 4. Convergence diagnostics from 4 dispersed chains of 10,000 draws. $\hat{R}$ is the larger of the rank-normalised and folded statistics, maximised over all 12 model parameters; ESS values are minima over the same parameters. The criterion is $\hat{R}$ below 1.01 and bulk ESS at least 400.

Sampler	$\hat{R}$	Bulk ESS	Tail ESS	Converged
MH	4.5314	4	11	no
Adaptive MH (naive)	3.4831	4	10	no
Preconditioned MH	1.0109	599	1059	no
Gibbs	1.0001	39068	37627	yes
HMC	1.0008	3670	8209	yes

Two samplers meet the criterion at the standard run length and three do not. Plain Metropolis reaches $\hat{R}$ 4.53 and adaptive Metropolis 3.48, both far above the threshold, with bulk ESS in single figures out of 40,000 draws. Neither has converged, and no quantity derived from them, whether a posterior mean, an efficiency figure or a predictive score, can be interpreted as an estimate of the target.

Preconditioned Metropolis does not converge at 10,000 draws per chain, with $\hat{R}$ 1.0109 exceeding the 1.01 threshold. Monitoring all 12 parameters exposed a coordinate that was not visible in a smaller diagnostic set. Extending to 40,000 draws per chain gives $\hat{R}$ 1.0033 and bulk ESS 2117, which does converge, confirming that the standard run length was insufficient rather than the method itself failing.

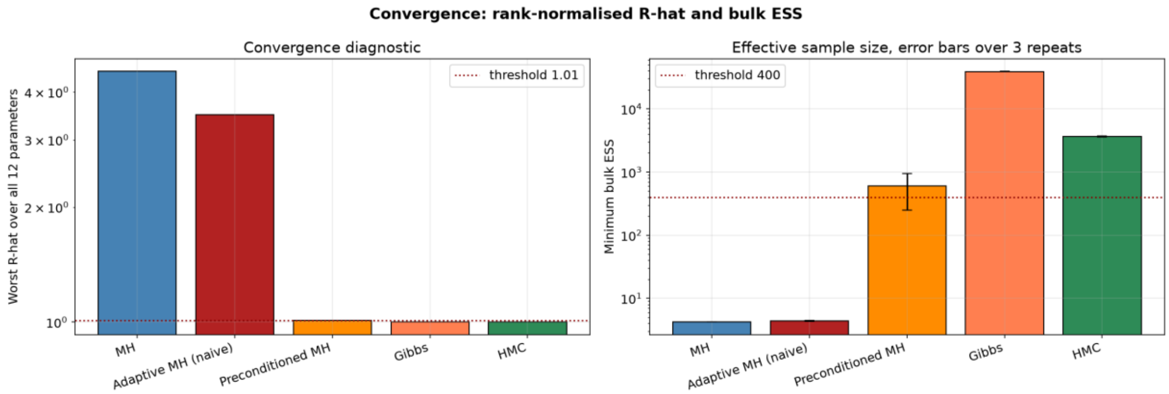

Figure 2. Worst-case R̂ and minimum bulk ESS for each sampler, both on logarithmic scales, with dashed lines marking the thresholds. Error bars on the ESS panel span the standard deviation across 3 independent repeats.

The consequences are visible in the estimates themselves. Compared with the deterministic quadrature reference of Section 5.5, the converged samplers agree to within 0.0010, while plain Metropolis is wrong by 1.36 and adaptive Metropolis by 1.44. For scale, the widest direction of the posterior has standard deviation 0.0499, so both errors exceed the entire spread of the posterior.

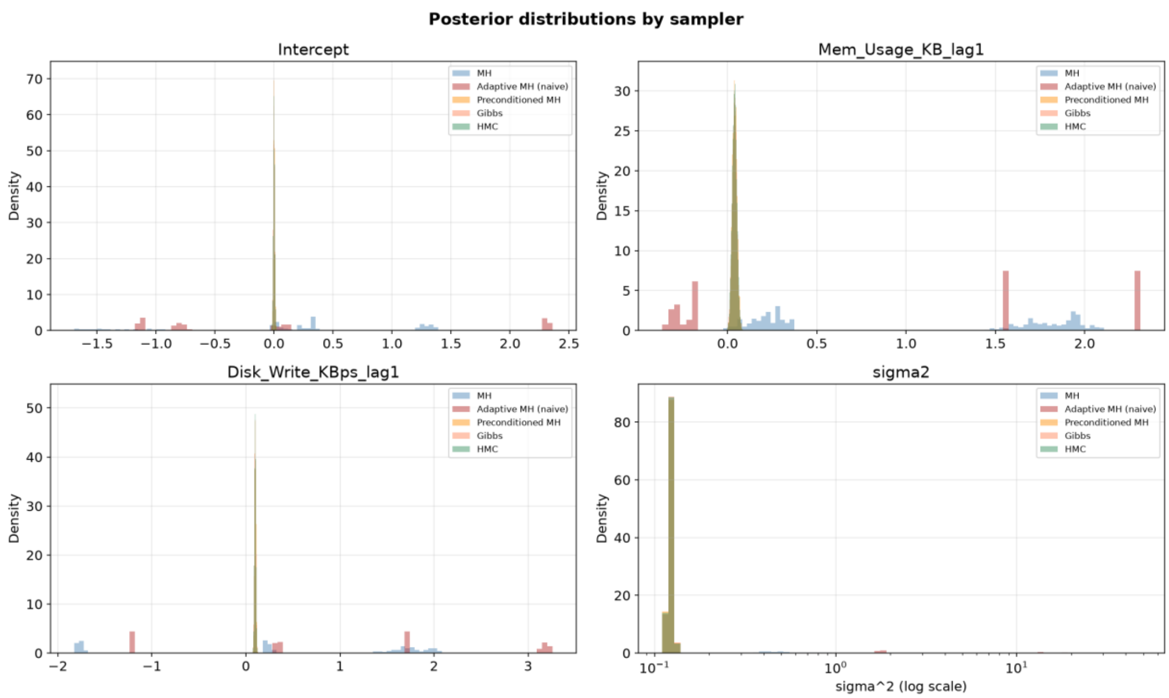

Figure 3. Posterior distributions by sampler for three coefficients and the noise variance. The converged samplers produce indistinguishable densities; the two that did not converge are displaced and too narrow.

5.3 Why Metropolis Fails, and What Repairs It

The failure of plain Metropolis follows from the geometry of Section 5.1 rather than from random-walk behaviour in the abstract. Two repairs were attempted and only one worked.

Estimating the proposal covariance from the chain's own history, the textbook adaptive recipe, made matters worse: acceptance collapsed to 1.1% and $\hat{R}$ rose to 3.48. The reason is a bootstrap problem. Adaptation needs the chain to have explored enough to estimate its own covariance, but the unpreconditioned chain has an integrated autocorrelation time of the order of the run length, so across 2,000 burn-in iterations it supplies only a handful of effectively independent points from which to estimate a 12-dimensional covariance. What the estimate captures instead is the transient drift from the dispersed start toward the mode, whose spread along the direction of travel is much larger than the posterior; scaling that by $2.38^2/d$ produces proposals so large that almost everything is rejected.

This should be read as a statement about this configuration, not about adaptive Metropolis in general. The method is standard and effective in many settings [5]. What this experiment shows is that it cannot bootstrap itself when the initial proposal mixes too slowly relative to the burn-in budget; a longer adaptation phase, or a better initial proposal, would be expected to rescue it.

Taking the metric from outside the chain does work, though the standard run length is not sufficient to demonstrate it. Preconditioning with the observed Fisher information brings acceptance to 24.6%, against the theoretical optimum of 23.4 percent, for essentially no additional cost per iteration (8.2 seconds against 14.7 for the full multi-chain run).

The bulk ESS of 599 recorded for the standard run is a nominal diagnostic improvement over the 4.2 of plain Metropolis, but it is not interpretable as a reliable efficiency figure: the chain did not meet the convergence criterion at that length, and an ESS computed from a chain that has not converged measures the rate at which it produces draws from the wrong distribution. The defensible comparison uses the extended run, which does converge. At 40,000 draws per chain preconditioned Metropolis reaches bulk ESS 2117, or 0.013 per retained draw.

Even on those valid numbers the gain should not be overstated. At 0.013 effective draws per retained draw, preconditioned Metropolis remains well behind Gibbs at 0.977. Matching the metric to the posterior removes the penalty for anisotropy; it does not remove the random walk.

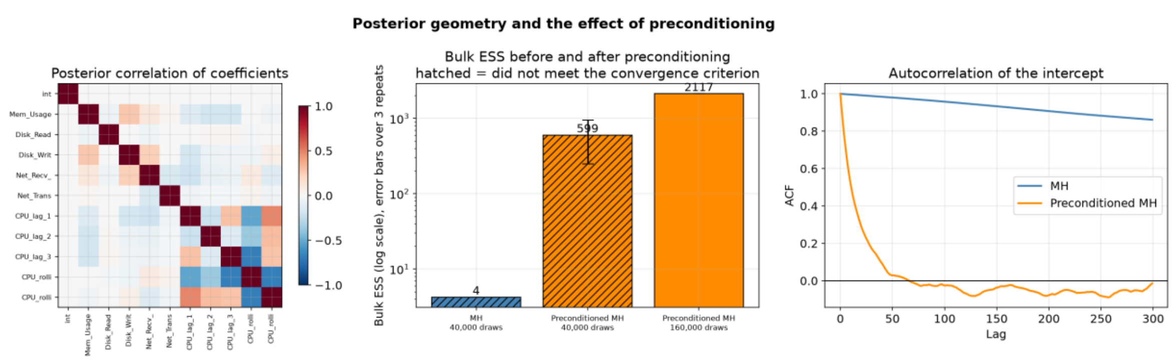

Figure 4. Left: posterior correlation between coefficients, showing the block of correlated lag and rolling features. Centre: bulk ESS before and after preconditioning, on a logarithmic scale; the preconditioned bar is from the standard run, which did not converge, so it is a diagnostic comparison rather than a certified efficiency figure. Right: autocorrelation of the intercept.

5.4 Efficiency

Table 5. Efficiency of the converged samplers. Runtime is the total for 4 chains of 10,000 draws, reported as mean and standard deviation over 3 repeats. ESS per second uses the same minimum bulk ESS and the same total runtime.

Sampler	Acceptance	Bulk ESS	Runtime (s)	Bulk ESS/s	MCSE (intercept)
Gibbs	N/A	39068	21.3 ± 3.6	1536	3.19e-05
HMC	95.1%	3670	91.8 ± 15.0	49	3.40e-05

Only the converged samplers are ranked. Including Metropolis or adaptive Metropolis in an efficiency table would invite a false comparison, because an ESS per second computed from a chain that is sampling the wrong distribution measures the rate at which it produces unusable draws. Their diagnostics appear in Table 4 and their runtimes in the accompanying results file, but they are excluded from every ranking in this section and from the predictive comparison in Section 6.

The ranking by bulk ESS per second is Gibbs (1536), HMC (49). Gibbs leads on both axes. Its worst-case bulk ESS is 0.977 per retained draw, so even the coordinate that mixes worst is close to independent. HMC converges under the selected configuration, but its worst-case ESS is 0.092 per retained draw, substantially below Gibbs in this conjugate model, and it additionally pays for 15 gradient evaluations per iteration. The Monte Carlo standard errors tell the same story in the units that matter for reporting a posterior mean.

5.5 Validation Against Independent References

Agreement among our own samplers is reassuring but limited, because all of them were written by the same authors from the same equations. Two independent references are therefore used, and they check different things.

An analytical reference.

Although the joint posterior has no standard closed form, β can be integrated out analytically: marginally, $y \mid \sigma^2 \sim N(0, \sigma^2 I + \tau^2 X X')$, whose log density is evaluated with Sylvester's determinant identity and the Woodbury identity. The remaining one-dimensional density over σ^2 is then integrated on a grid of 6,000 points, and the posterior mean of β recovered as the corresponding mixture of conditional means. This reference involves no sampling whatsoever, so it validates the transcription of the likelihood and the priors, not merely our sampling of them. Gibbs reproduces it to within 0.00014 and the reference posterior mean of σ^2 is 0.12273.

An external sampler.

The same posterior was also sampled with emcee 3.1.6 [14], which implements the affine-invariant ensemble algorithm of Goodman and Weare [15]. It shares nothing with our methods: no conditional structure, no gradients and no proposal covariance, and it is written by others. Running 40 walkers for 12,000 steps and discarding 4,000 produced 320,000 draws in 58.0

seconds, with worst R̂ 1.0229 and minimum bulk ESS 1989. Its posterior means agree with Gibbs to within 0.00104.

The scope of this second check should be stated precisely. Agreement with emcee provides independent evidence that our custom samplers correctly target the supplied log-posterior. Because emcee evaluates the same user-defined log-posterior function, the comparison does not independently validate the transcription of the likelihood and the priors. That is the role of the analytical reference above, which is constructed from the model algebra rather than from our code path.

One caveat: at R̂ 1.0229 the emcee run is itself slightly above the 1.01 threshold used for our own samplers. Ensemble samplers correlate their walkers by construction, so the split-chain diagnostic is conservative when applied to them, but the agreement it supports is correspondingly approximate and is quoted here to four decimal places rather than more.

The No-U-Turn Sampler was not used as a reference; no NUTS results are reported in this study.

5.6 Sensitivity to Tuning Parameters

Metropolis and HMC both require tuning; the Gibbs samplers do not. For HMC the step size ε and the number of leapfrog steps L are varied jointly, because they trade off against one another and neither is interpretable alone. Efficiency is reported as bulk ESS per gradient evaluation, which is the natural currency for a gradient method, alongside acceptance rate and R̂.

The grid runs differ from the main experiment in one respect that should be stated. They are warm started at the least-squares fit with small random perturbations, rather than from the overdispersed points used in Section 5.2. Short runs from dispersed starts would spend most of their length travelling to the posterior, so the resulting diagnostics would measure burn-in rather than the effect of the tuning parameters, which is what the grid is meant to isolate. The cost of that choice is that split R̂ is a weaker guarantee here than in Table 4, because chains that begin close together can agree without having explored the posterior. The grid should therefore be read as ranking configurations against one another, not as certifying any of them as converged. Each cell runs 4 chains with 1,500 burn-in iterations and 4,000 retained draws per chain.

Table 6. HMC sensitivity to step size and trajectory length, from 4 warm-started chains with 1,500 burn-in iterations and 4,000 retained draws each. The final column records whether a cell meets the diagnostic thresholds of Section 2.7; because the chains are warm started, meeting them ranks a configuration rather than certifying convergence. ESS values for cells below the thresholds are not interpretable and are shown only for completeness.

ε	L	Acceptance	R̂	Bulk ESS	ESS/grad.	Meets warm-start thresholds
0.001	5	98.8%	1.1251	38	2.90e-04	no
0.001	10	99.2%	1.0272	195	8.07e-04	no
0.001	15	98.9%	1.0119	405	1.15e-03	no

ε	L	Acceptance	$\hat{R}$	Bulk ESS	ESS/grad.	Meets warm-start thresholds
0.001	20	99.2%	1.2814	729	1.58e-03	no
0.002	5	96.8%	1.0282	181	1.37e-03	no
0.002	10	96.3%	1.2864	692	2.86e-03	no
0.002	15	95.1%	1.0024	1582	4.49e-03	yes
0.002	20	95.7%	1.1540	23	4.97e-05	no
0.004	5	78.0%	1.1845	551	4.17e-03	no
0.004	10	93.0%	1.1419	23	9.46e-05	no
0.004	15	78.3%	1.0121	5439	1.55e-02	no
0.004	20	90.6%	1.0490	107	2.33e-04	no
0.008	5	0.0%	10.7330	4	3.07e-05	no
0.008	10	0.0%	degenerate	4	1.65e-05	no
0.008	15	0.0%	degenerate	4	1.14e-05	no
0.008	20	0.0%	degenerate	4	8.67e-06	no

One of the 16 configurations met the diagnostic thresholds in the warm-start grid, and the comparison is restricted to it. The best-performing configuration among those meeting the warm-start diagnostic thresholds is ε = 0.002 with L = 15, at 4.49e-03 bulk ESS per gradient evaluation and an acceptance rate of 95.1%.

This directly contradicts the reading of a high acceptance rate as a sign of good tuning. The highest acceptance in the grid, 99.2% at ε = 0.001 and L = 10, is not the most efficient configuration; a step size that small buys near-certain acceptance by proposing moves so short that the trajectory barely advances, and the gradient budget is spent for little gain. The theoretically motivated target for HMC is roughly 65 to 80 percent [16]. Every configuration with a step size of ε = 0.002 or smaller, which includes the ε = 0.002 used elsewhere in the study, has an acceptance rate of at least 95.1%, far above that target, indicating that the step size used elsewhere is conservative. At the other extreme, ε = 0.008 breaks the leapfrog integrator entirely and acceptance falls to zero.

Table 7. Metropolis acceptance against proposal scale, at fixed scale 0.05 for log σ^2.

Proposal scale for β	Acceptance rate
0.0005	49.7%
0.0010	48.5%
0.0050	46.6%
0.0100	35.9%
0.0200	14.8%
0.0500	3.8%

The Metropolis grid shows the familiar trade-off, but it is worth reading together with Section 5.1: no scalar step size performs well, because the search is over a one-dimensional family for a problem that requires a matrix.

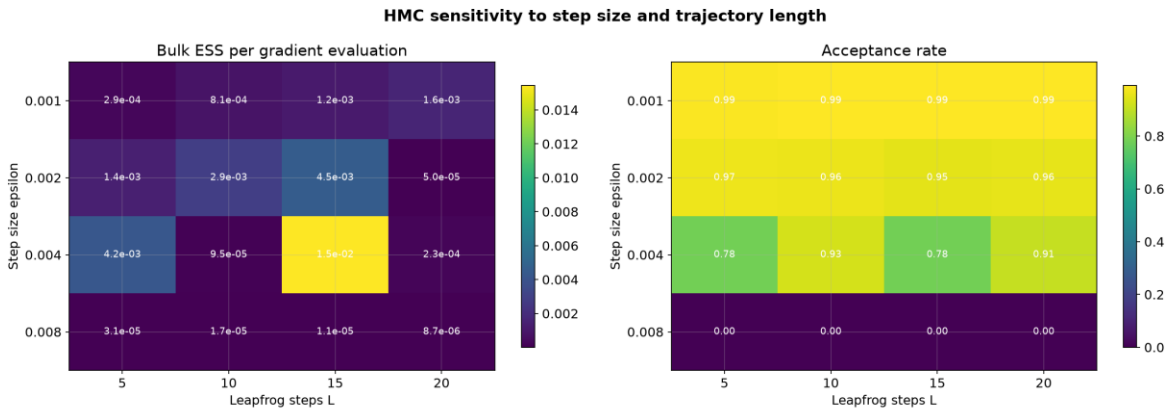

Figure 5. Bulk ESS per gradient evaluation (left) and acceptance rate (right) over the grid of step sizes and trajectory lengths. The most efficient configuration is not the one with the highest acceptance.

5.7 Sensitivity to the Starting Point

MCMC converges from any starting point in theory but says nothing about how long that takes. Each sampler was started from four deliberately different points and run for 1,500 iterations. The median and median absolute deviation (MAD) of the log posterior are taken over the final 500 iterations. A chain is recorded as having entered the stable region at the first iteration from which its log posterior stays within 3 MADs of that median for 50 consecutive iterations. The sustained window is what makes this measure the end of the initial transient: a rule that instead demanded containment for the whole remainder of the run would report the last random excursion out of the band, which happens late in a chain that is already mixing well and has nothing to do with the starting point.

This is a descriptive measure of when the starting point stops mattering. It is not evidence of stationarity, and it is not used anywhere in this report as a convergence criterion; that role belongs to the $\hat{R}$ and ESS diagnostics of Section 5.2.

Table 8. Iterations after which the log posterior stays within 3 MADs of its final-500-iteration median for 50 consecutive iterations, by sampler and starting point.

Sampler	zeros	least squares fit	dispersed (+3)	extreme variance
Preconditioned MH	384	67	497	320
Gibbs	20	20	20	20
HMC	284	284	284	284

Ranked by the slowest of the four starts, the order is Gibbs at 20 iterations, HMC at 284 iterations, Preconditioned MH at 497 iterations. Gibbs is insensitive to where it starts because its first draw of β comes from the exact conditional given σ^2 and therefore lands in the posterior bulk immediately, so there is almost no transient to measure. The ordering otherwise follows how far each sampler moves per iteration: an exact conditional draw crosses the posterior in one step, a Hamiltonian trajectory in a few, and a preconditioned random walk in many. These counts are consistent with Figure 6, where the traces from different starts become visually indistinguishable at roughly the iterations marked by the dotted lines.

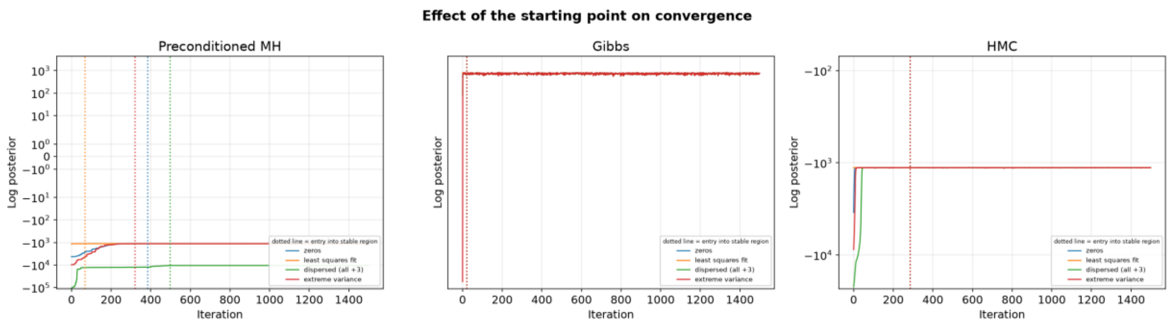

Figure 6. Log posterior over the full 1,500 iterations from four starting points, on a symmetric logarithmic scale. The dotted vertical line in each colour marks where that start enters the stable region, as tabulated in Table 8. Curves that meet quickly indicate the starting point has been forgotten.

6. Predictive Performance and Calibration

6.1 Construction of Posterior Predictive Intervals

All intervals reported in this section are posterior predictive intervals for a future CPU observation, not credible intervals for a parameter. The distinction matters because the two have different widths and answer different questions. For each retained posterior draw (β_s, σ^2_s) a replicate observation is generated as

$$y^* = x'\beta_s + \sigma_s e, \quad e \sim N(0, 1) \ \text{or} \ e \sim t_\nu$$

so each replicate carries both sources of uncertainty: parameter uncertainty, through the spread of β_s and σ_s across draws, and observation noise, through e. The interval is then the empirical quantile range of these replicates, rather than a point estimate plus a multiple of a standard deviation, so no symmetry or normality is imposed on the result.

6.2 Selecting the Likelihood on Validation Data

The choice between the Gaussian and Student-t likelihoods, and the value of ν, is made on the validation split alone, by mean log pointwise predictive density. The predictive density is the posterior mixture $(1/S) \sum_s p(y^* \mid \theta_s)$, and each component is evaluated in closed form, so the criterion does not depend on simulated replicates.

Table 9. Likelihood selection on the validation split. Higher log predictive density is better; nominal coverage is 50 and 95 percent. The selected row is the Student-t with ν = 2.

Candidate	Log predictive density	Coverage 50%	Coverage 95%	Median abs. error
Gaussian	-0.3862	94.2%	98.4%	0.0433
Student-t, ν = 2	1.6305	74.9%	97.7%	0.0143

Candidate	Log predictive density	Coverage 50%	Coverage 95%	Median abs. error
Student-t, v = 3	1.6158	74.4%	97.5%	0.0162
Student-t, v = 4	1.5898	74.9%	97.1%	0.0170
Student-t, v = 6	1.5333	75.0%	96.7%	0.0182
Student-t, v = 10	1.4258	75.9%	96.8%	0.0197
Student-t, v = 20	1.2062	78.2%	97.4%	0.0214

The criterion prefers the Student-t at every value of v tested, and prefers heavier tails monotonically, selecting the smallest value on the grid, v = 2. Two qualifications follow. First, v = 2 has infinite variance, so the fitted model has a well-defined median and scale but no finite predictive variance; this is defensible for a criterion based on predictive density but would be inappropriate if a variance were needed downstream. Second, because the selected value sits at the edge of the grid, the data are indicating that they would prefer heavier tails still, and a wider grid or a prior on v would be the principled next step rather than accepting an edge solution. The sensitivity of the criterion across the grid is shown below; the ordering is stable and not an artefact of a single value.

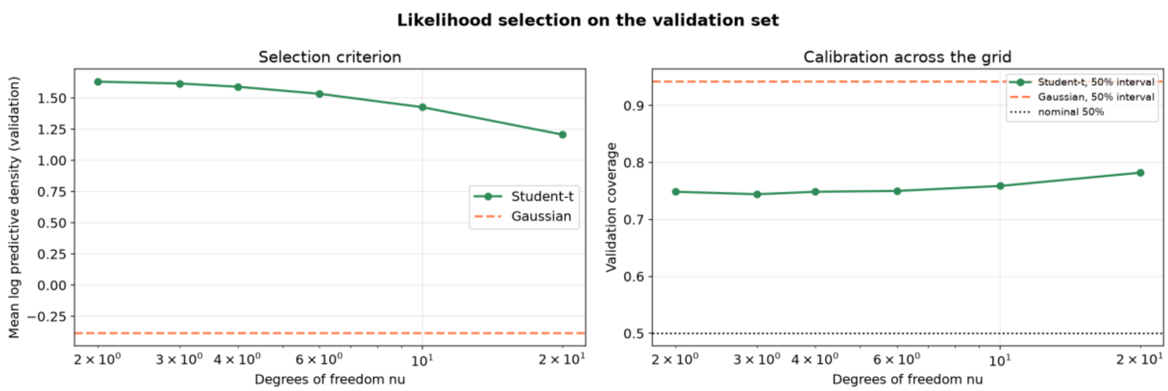

Figure 7. Validation log predictive density (left) and 50 percent interval coverage (right) across the grid of degrees of freedom, with the Gaussian shown as a horizontal reference. The criterion decreases monotonically in v.

6.3 Test-Set Performance

The test split is used here and nowhere else. Errors are reported both in standardised units and in CPU percentage points, obtained by multiplying by the training standard deviation of the target, 4.999 percentage points.

The Student-t model is held to the same standard as the samplers in Section 5. It is fitted with 4 chains from overdispersed starting points and diagnosed with the same estimators and the same worst-case aggregation, giving $\hat{R}$ 1.0013 and minimum bulk ESS 2272, which meets the criterion of $\hat{R}$ below 1.01 and bulk ESS at least 400 used throughout. Every Student-t number reported in this section comes from those 4 chains pooled, not from a single run. This matters because the argument of Section 5 is that a sampler which has not been diagnosed cannot support conclusions, and that argument applies to the model used for the predictive results just as much as to the ones being compared.

Table 10. Test-set accuracy in standardised units and in CPU percentage points.

Model	RMSE	RMSE (pp)	Median abs. error	Median abs. error (pp)
OLS baseline	0.2658	1.329	0.0421	0.210
Bayesian, Gaussian likelihood	0.2658	1.329	0.0420	0.210
Bayesian, Student-t (ν = 2)	0.2047	1.023	0.0145	0.072

The Gaussian Bayesian model and ordinary least squares are indistinguishable in point accuracy, which is expected: with weak priors and 3,000 training observations the posterior mean is essentially the least-squares solution, and the value of the Bayesian treatment lies in the predictive distribution rather than the point estimate. The gap between RMSE and median absolute error, a factor of about 6, is the first sign that the error distribution is far from Gaussian: a few large errors dominate the squared measure while the typical error is much smaller. The Student-t model reduces the median absolute error from 0.210 to 0.072 percentage points.

6.4 Residual Behaviour

Two properties of the residuals shape the calibration results. They are strongly non-Gaussian, with excess kurtosis 111 and a ratio of standard deviation to robust standard deviation of 5.12, so almost all residuals are small and a few are very large. This is what a CPU trace of quiet periods punctuated by spikes produces. A Gaussian likelihood has one parameter with which to describe both regimes and inflates σ^2 to cover the spikes.

They also remain autocorrelated within each machine. The median per-machine lag-one residual autocorrelation is -0.201 and 15 of 48 machines reject the Ljung-Box test at the 5 percent level (median $p = 0.093$). The model treats observations as conditionally independent given the predictors, and the lag terms absorb much but not all of the serial structure within each machine. Remaining autocorrelation means the effective number of independent test observations is smaller than the nominal count, which is why the coverage estimates below are accompanied by cluster-bootstrap intervals that resample machines with replacement rather than binomial ones.

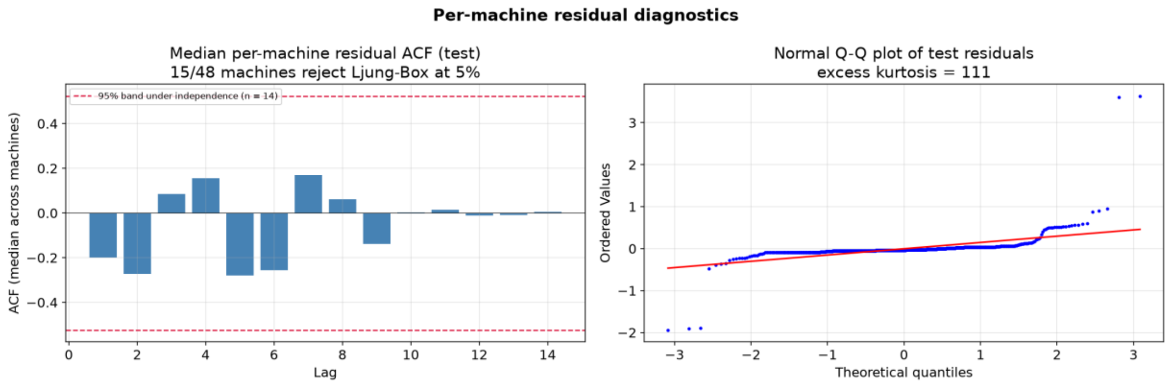

Figure 8. Left: median per-machine residual autocorrelation on the test split. The dashed lines mark the 95 percent band expected under independence, 1.96 divided by the square root of the median per-machine series length of 14 observations. Right: normal Q-Q plot of the test residuals, whose S-shape is the signature of heavy tails.

6.5 Calibration

Table 11. Posterior predictive interval coverage on the test split, with cluster-bootstrap 95 percent intervals (2,000 replicates, resampling machines with replacement) that account for within-machine serial dependence. Widths are in CPU percentage points.

Nominal	Gaussian	Gaussian 95% CI	Student-t	Student-t 95% CI	Width Gauss (pp)	Width t (pp)
50%	94.7%	[88.9%, 99.0%]	74.1%	[68.1%, 79.6%]	2.36	0.33
95%	98.8%	[96.6%, 100.0%]	97.6%	[94.7%, 99.6%]	6.86	1.75

Under the Gaussian likelihood the intervals are far too wide. A nominal 50 percent interval covers 94.7% of test observations. Coverage is 44.7 percentage points above the nominal 50 percent level, and the nominal 95 percent interval covers 98.8%. The bootstrap intervals show this is not sampling noise.

The Student-t likelihood improves calibration substantially but does not fix it. Coverage of the nominal 50 percent interval falls from 94.7% to 74.1% and the mean width from 2.36 to 0.33 percentage points, yet 74.1% against a nominal 50 percent is still severely miscalibrated: coverage is 24.1 percentage points above the nominal 50 percent level. This is a partial improvement, not a repair, and the residual miscalibration is consistent with the diagnostics of Section 6.4: a Student-t with a single scale still assumes one noise level for every machine and every time, whereas the data are heteroscedastic across machines, as Section 3.5 showed, and serially dependent.

A useful check confirms that the cause lies in the likelihood rather than in the sampler or the priors: the same over-coverage appears when intervals are built from an ordinary least-squares fit and its residual standard deviation, with no Bayesian machinery at all.

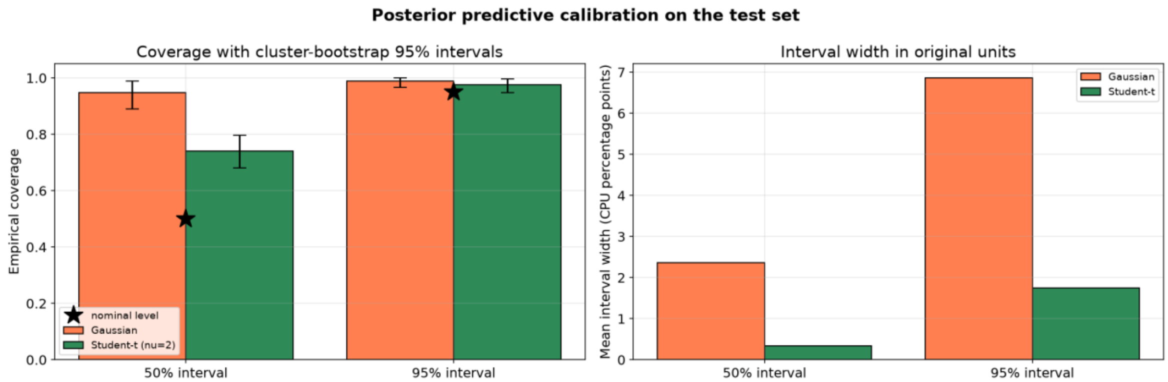

Figure 9. Left: coverage of posterior predictive intervals under each likelihood, with cluster-bootstrap intervals and stars marking the nominal levels. Right: mean interval width in CPU percentage points.

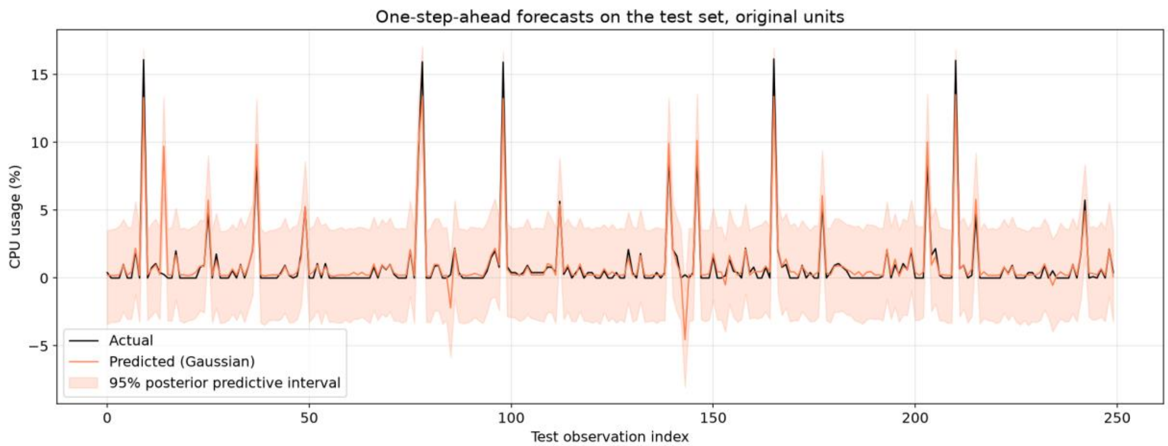

Figure 10. One-step-ahead forecasts in original units, with the 95 percent posterior predictive interval, for the first 250 of the 680 test observations.

7. Discussion

Answers to the research questions.

RQ1. Two of the five samplers converge at the standard 10,000 draws per chain. Preconditioned Metropolis converges only with a longer run of 40,000 draws. The two that fail entirely do so because the posterior is anisotropic, with condition number 314 arising from the near-collinear lag features, and because an isotropic random-walk proposal cannot cope with that geometry. Supplying a metric from the observed Fisher information largely repairs it; estimating the metric from the chain itself does not, under the burn-in budget tested.

RQ2. Among the samplers that converge at the standard run length, Gibbs is the most efficient both per second and per draw in this conjugate setting, at 0.977 worst-case bulk ESS per retained draw. HMC converges under the selected configuration, but its worst-case ESS is 0.092 per retained draw, substantially below Gibbs, and it also pays for L gradient evaluations per iteration. The sensitivity grid shows its efficiency per gradient depends jointly on step size and trajectory length, with the best-performing configuration not the one with the highest acceptance rate.

RQ3. The intervals are not calibrated. A Gaussian likelihood produces nominal 50 percent intervals covering 94.7%, because heavy-tailed residuals inflate the single noise parameter. A Student-t likelihood selected on validation data reduces this to 74.1%, a substantial but partial improvement.

Strengths and weaknesses of each sampler.

- Metropolis-Hastings: the most general, needing only pointwise density evaluation, and the cheapest per iteration. On this posterior it did not converge, so its generality is of no practical use here without preconditioning.
- Adaptive Metropolis: sound in principle and standard in practice, but unable to bootstrap a proposal covariance from a chain that mixes too slowly during the burn-in available. A longer adaptation phase would be the obvious remedy.
- Preconditioned Metropolis: keeps the generality of Metropolis while removing the anisotropy penalty, at negligible extra cost, and needs no conjugacy. It converges only with a longer run and remains a random walk, an order of magnitude behind Gibbs and HMC in effective sample size.
- Gibbs: no tuning, exact conditionals, insensitive to the starting point, and the fastest here. It requires closed-form conditionals, which exist only because of the prior structure chosen, so it does not generalise to most realistic models.
- Hamiltonian Monte Carlo: converges reliably and is expected to scale better to higher dimensions than a random walk, at the cost of a differentiable log posterior and two tuning parameters. On this posterior its worst-case ESS is 0.092 per retained draw, well below the 0.977 achieved by Gibbs, so the conjugate structure that Gibbs exploits is worth more here than gradient information.

Limitations.

- The model is linear and pools all machines under one coefficient vector, which Section 3.5 shows is only approximately right.
- Only 4,368 observations are used, so the test split contains 680 rows and the coverage estimates carry the uncertainty shown by the bootstrap intervals.
- The selected degrees of freedom sits at the edge of the tested grid, so the data would prefer heavier tails than the grid allows.
- Residual autocorrelation remains, so the conditional independence assumption of the likelihood is not satisfied and reported uncertainties are optimistic in that respect.
- Conclusions come from one dataset, and the ranking of samplers depends on posterior geometry, which is dataset-specific.

8. Conclusions

- On this posterior, Gibbs sampling and Hamiltonian Monte Carlo converge at the standard 10,000 draws per chain, reproducing a deterministic quadrature reference to within 0.0010. Fisher-preconditioned Metropolis converges only with a longer run of 40,000 draws. Random-walk Metropolis and adaptive Metropolis do not converge, reaching $\hat{R}$ 4.53 and 3.48.
- The failure is explained by posterior anisotropy, condition number 314, caused by near-collinear lag features. Preconditioning with the observed Fisher information repairs the geometry at negligible cost: extended to 40,000 draws per chain it converges and reaches bulk ESS 2117, against 4.2 for plain Metropolis.
- Estimating the proposal covariance from the chain itself failed under the configuration tested, collapsing acceptance to 1.1%. This is a property of the burn-in budget and initial proposal, not a general property of adaptive Metropolis.
- For HMC, efficiency per gradient evaluation depends on step size and trajectory length jointly. The best-performing configuration among those meeting the warm-start diagnostic thresholds, $\varepsilon = 0.002$ with $L = 15$, has an acceptance rate of 95.1%, lower than several less efficient settings, so a high acceptance rate is not by itself evidence of good tuning.
- Posterior predictive intervals from a Gaussian likelihood are badly miscalibrated, a nominal 50 percent interval covering 94.7%, because test residuals have excess kurtosis 111. A Student-t likelihood chosen on validation data improves this to 74.1% and reduces the median absolute error from 0.210 to 0.072 percentage points, though coverage remains 24.1 percentage points above the nominal level.

Further work.

- Place a prior on ν and sample it, rather than selecting from a grid whose edge is chosen.
- Fit a hierarchical model with per-machine coefficients and variances, which would address both the pooling limitation and the residual heteroscedasticity.
- Model the spike regime explicitly, for instance with a mixture or a state-space component, instead of absorbing it into heavy tails.
- Implement the No-U-Turn Sampler, which would remove the trajectory-length tuning that Section 5.6 shows matters.

9. References

[1] Shen, S., van Beek, V., & Iosup, A. (2015). Statistical characterization of business-critical workloads hosted in cloud datacenters. In Proceedings of the 15th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), 465-474. https://doi.org/10.1109/CCGrid.2015.60. Dataset: Grid Workloads Archive GWA-T-12 Bitbrains, http://gwa.ewi.tudelft.nl/datasets/gwa-t-12-bitbrains

[2] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Burkner, P. C. (2021). Rank-normalization, folding, and localization: an improved R-hat for assessing convergence of MCMC. Bayesian Analysis, 16(2), 667-718. https://doi.org/10.1214/20-BA1221

[3] Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., & Teller, E. (1953). Equation of state calculations by fast computing machines. The Journal of Chemical Physics, 21(6), 1087-1092. https://doi.org/10.1063/1.1699114

[4] Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their applications. Biometrika, 57(1), 97-109. https://doi.org/10.1093/biomet/57.1.97

[5] Haario, H., Saksman, E., & Tamminen, J. (2001). An adaptive Metropolis algorithm. Bernoulli, 7(2), 223-242. https://doi.org/10.2307/3318737

[6] Geman, S., & Geman, D. (1984). Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. IEEE Transactions on Pattern Analysis and Machine Intelligence, 6(6), 721-741. https://doi.org/10.1109/TPAMI.1984.4767596

[7] Neal, R. M. (2011). MCMC using Hamiltonian dynamics. In Handbook of Markov Chain Monte Carlo (pp. 113-162). CRC Press. https://doi.org/10.1201/b10905

[8] Roberts, G. O., Gelman, A., & Gilks, W. R. (1997). Weak convergence and optimal scaling of random walk Metropolis algorithms. The Annals of Applied Probability, 7(1), 110-120. https://doi.org/10.1214/aoap/1034625254

[9] West, M. (1984). Outlier models and prior distributions in Bayesian linear regression. Journal of the Royal Statistical Society: Series B, 46(3), 431-439. https://doi.org/10.1111/j.2517-6161.1984.tb01317.x

[10] Lange, K. L., Little, R. J. A., & Taylor, J. M. G. (1989). Robust statistical modeling using the t distribution. Journal of the American Statistical Association, 84(408), 881-896. https://doi.org/10.2307/2290063

[11] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). Bayesian Data Analysis (3rd ed.). CRC Press.

[12] Geyer, C. J. (1992). Practical Markov Chain Monte Carlo. Statistical Science, 7(4), 473-483. https://doi.org/10.1214/ss/1177011137

[13] Kumar, R., Carroll, C., Hartikainen, A., & Martin, O. (2019). ArviZ: a unified library for exploratory analysis of Bayesian models in Python. Journal of Open Source Software, 4(33), 1143. https://doi.org/10.21105/joss.01143

[14] Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. (2013). emcee: The MCMC Hammer. Publications of the Astronomical Society of the Pacific, 125(925), 306-312. https://doi.org/10.1086/670067

[15] Goodman, J., & Weare, J. (2010). Ensemble samplers with affine invariance. Communications in Applied Mathematics and Computational Science, 5(1), 65-80. https://doi.org/10.2140/camcos.2010.5.65

[16] Beskos, A., Pillai, N., Roberts, G., Sanz-Serna, J. M., & Stuart, A. (2013). Optimal tuning of the hybrid Monte Carlo algorithm. Bernoulli, 19(5A), 1501-1534. https://doi.org/10.3150/12-BEJ414

10. Code and Reproducibility

All code and instructions are available at:

https://github.com/eladagmi24/mcmc-sampling-project

Table 12. Repository contents.

Path	Contents
src/run_experiment_v2.py	All samplers, diagnostics, references, studies, figures
src/mcmc_diagnostics.py	Rank-normalised $\hat{R}$ and bulk/tail ESS, from scratch
src/create_report_v2.py	Generates this document from the results file
src/finalise_document.ps1	Evaluates the contents field and exports a PDF
results/experiment_results_v2.json	Every number appearing in this document
results/figures/	All figures
notebooks/	Annotated notebook
data/	Not committed; see data/README.md for the download link and layout

The pipeline is reproduced with two commands, plus an optional third that asks Word to evaluate the table-of-contents field and export a PDF:

```
python src/run_experiment_v2.py --skip-nuts
python src/create_report_v2.py
powershell -File src/finalise_document.ps1
```

The Bitbrains traces are not committed: the fastStorage directory is 1.19 GB across 1,250 files and is published by its authors [1]. The --skip-nuts flag omits an optional PyMC cross-check that is impractically slow without a C++ compiler; it is not used anywhere in this document, and the external validation of Section 5.5 uses emcee instead.